RESEARCH ARTICLE

Vasopressin increases human risky cooperative behavior

Claudia Brunnlieb, Gideon Nave, Colin F. Camerer, +3, and Marcus Heldmann Authors Info & Affiliations

Edited by Michael S. Gazzaniga, University of California, Santa Barbara, CA, and approved January 5, 2016 (received for review September 24, 2015)

February 8, 2016 113 (8) 2051-2056 https://doi.org/10.1073/pnas.1518825113

12,683 | 56

Significance

Most forms of cooperative behavior take place in a mutually beneficial context where cooperation is risky as its success depends on unknown actions of others. In two pharmacological experiments, we show that intranasal administration of arginine vasopressin (AVP), a hormone that regulates mammalian social behaviors such as monogamy and aggression, increases humans' tendency to engage in mutually beneficial cooperation. Several control tasks ruled out that AVP's effects were driven by increased willingness to bare risks in the absence of social context, beliefs about the actions of one's partner, or altruistic concerns. Our findings provide novel causal evidence for a biological factor underlying cooperation and are in accord with previous findings that cooperation is intrinsically rewarding for humans.

Abstract

The history of humankind is an epic of cooperation, which is ubiquitous across societies and increasing in scale. Much human cooperation occurs where it is risky to cooperate for mutual benefit because successful cooperation depends on a sufficient level of cooperation by others. Here we show that arginine vasopressin (AVP), a neuropeptide that mediates complex mammalian social behaviors such as pair bonding, social recognition and aggression causally increases humans' willingness to engage in risky, mutually beneficial cooperation. In two double-blind experiments, male participants received either AVP or placebo intranasally and made decisions with financial consequences in the "Stag hunt" cooperation game. AVP increases humans' willingness to cooperate. That increase is not due to an increase in the general willingness to bear risks or to altruistically help others. Using functional brain imaging, we show that, when

We use cookies on this site to enhance your user experience. By using this website, you are giving your consent for us to set cookies. Find out more

CONTINUE

dlPFC functional connectivity with the ventral pallidum, an AVP receptor-rich region previously associated with AVP-mediated social reward processing in mammals. These findings show a previously unidentified causal role for AVP in social approach behavior in humans, as established by animal research.

Sign up for PNAS alerts.

Get alerts for new articles, or get an alert when an article is cited.

MANAGE ALERTS

No other species shows the level of cooperative behavior that humans do. From hunter–gatherer communal sharing to spice trading to international response to catastrophes, an ever-increasing scale of cooperation marks the history of humankind and accounts for its successes (1). How have humans evolved to cooperate? Cooperation is often studied using behavioral paradigms where it is costly for an individual to cooperate, such as in social dilemmas in which prosocial choices benefit another individual at a cost to oneself (2). However, naturally occurring cooperation is often individually beneficial provided enough other individuals are also cooperative (called "risky cooperation") (3, 4). A recent theory, the "interdependent hypothesis," postulates that humans have evolved a unique skill for mutually beneficial cooperation in the context of especially risky coordination problems (5). The interdependent hypothesis is consistent with the risk sensitivity observed in primate group hunting in the wild (6, 7) with recent comparative studies reporting that humans are more cooperative than other primates only when cooperation requires increased willingness to bear social risks (8). As human cooperative behavior is associated with self-reported measures of pleasure and satisfaction (9, 10) and neural activation of the reward system (11, 12), it seems that our species has developed a biological mechanism that allows us to overcome the riskiness of cooperation to coordinate mutually beneficial social actions (13).

The development of neuromodulatory systems in the brain is a candidate biological mechanism underlying the evolution of mammalian social behaviors (14). Specifically, the neuropeptide arginine vasopressin (AVP) acts both as a neurotransmitter and a hormone with widespread targets including regions of the prefrontal cortex, the amygdala, and the hippocampus and interacts with dopaminergic, reward-processing circuits in the ventral pallidum (VP) and nucleus accumbens (14). AVP regulates male mammalian social behaviors such as affiliation, aggression, monogamy, and paternal behaviors (14). Scholars have suggested that AVP mediates male pair bonding in mammals by stimulating dopamine release in the VP, inducing reward associations with the partner (15). Rodent studies have also linked AVP with social recognition, an important capacity required for distinguishing between conspecifics (16).

We use cookies on this site to enhance your user experience. By using this website, you are giving your consent for us to set cookies. Find out more

CONTINUE

Like cooperation, many AVP-mediated mammalian social behaviors are potentially rewarding but are also risky: the benefits of spending all mating resources on a single female partner and engaging in paternal investment (at the absence of assured paternity), for example, depend on uncertain actions of con-specifics (17). Based on the proposition that AVP enhances the reward associated with social interactions in mammals (18), we hypothesized that AVP encodes the intrinsic value associated with cooperation in humans, overcoming the riskiness of social interactions to facilitate cooperation among humans.

AVP crosses the blood–brain barrier after intranasal administration, providing means for directly manipulating the level of AVP in the human central nervous system (19). Here we delineate the impact of intranasal AVP on risky cooperative behavior in a double-blind, placebo-controlled design, using a Stag hunt game, an experimental economic paradigm that is commonly used to study mutually beneficial risky cooperation (20, 21) (Fig. 1). In the Stag hunt, two players simultaneously choose between a cooperative action that potentially yields a high payoff (A, "Stag") and a noncooperative action (B, "Rabbit"), the payoffs of which are lower but less risky (the choice labels "Stag" and "Rabbit" originate from a stylized description of the choice between safe individual hunting for low-value Rabbits, compared with hunting for a high-value Stag that requires many individuals to collaborate in the hunt). By choosing Stag, a player may increase his or her payoff, but also risks ending up worse off if his partner did not cooperate by choosing Stag.

Fig. 1.

A

	stag	rabbit
stag	€200 €200	€0 €90
rabbit	€90 €0	€160 €160

B

	stag	rabbit
stag	€200 €200	€0 *S*
rabbit	*S* €0	€160 €160

The Stag hunt task. (A) Stag hunt game payoff matrix. The cooperative action Stag potentially generates the highest possible payoff, but only if one's partner cooperates as well. Rabbit choice secures a lower (but certain) payoff, regardless of the partner's action. The payoff dominant equilibrium (200/200) is at the Top Left, and the risk-dominant equilibrium (160/160) is at the Bottom Right. (B) Stag hunt game payoff matrix of the strategy method. Participants indicated a security level (parameter S-value) at which they would switch between the risk-dominant Rabbit strategy to the payoff-dominant Stag strategy and also indicated the direction of switching.

The Stag hunt game has two stable behavioral patterns that are mutually optimal for both players (called "pure Nash equilibria" in game theory). If one's partner is sufficiently likely to choose Stag, then Stag choice is optimal for an individual. The (Stag, Stag) choice pair is therefore a self-fulfilling equilibrium pattern and earns the highest payoffs for both players (Fig. 1*A*, top left of the payoff matrix). In the other equilibrium pattern, both players fear that the partner will not cooperate and hence choose the safer Rabbit strategy (Fig. 1*A*, bottom right of the payoff matrix). Note that (Stag, Stag) groups earn more, collectively, than (Rabbit, Rabbit)

We use cookies on this site to enhance your user experience. By using this website, you are giving your consent for us to set cookies. Find out more
CONTINUE

beneficial social risk-taking. Crucially, the Stag hunt game is different from a social dilemma

game (e.g., the prisoner's dilemma or the public goods game) because even in a single play of the game, the cooperative Stag choice is individually optimal if one's partner is likely to choose Stag (on the contrary, defection is the optimal strategy of a selfish player facing a dilemma, regardless of the partner's behavior).

We manipulated the social risk associated with a Stag hunt game by varying its *security level* (denoted "*S*"). The security level *S* is the minimal payoff a player could secure himself by choosing Rabbit (Fig. 1*B*). A high security level makes the Rabbit action more attractive; thus, Stag becomes less attractive and more risky, as the likelihood that one's partner will choose Stag presumably decreases. We hypothesized that AVP will increase the rates of Stag choice in the Stag hunt game and also tested whether any effect of AVP is due to other measurable factors that could potentially influence Stag choices, namely, subjects' general attitude toward financial risk, the belief about the partners' strategy choice, and social preferences (i.e., willingness to incur a cost to benefit others). Finally, we investigate the neural mechanisms underlying AVP's effects on human cooperative behavior using functional magnetic resonance imaging (fMRI).

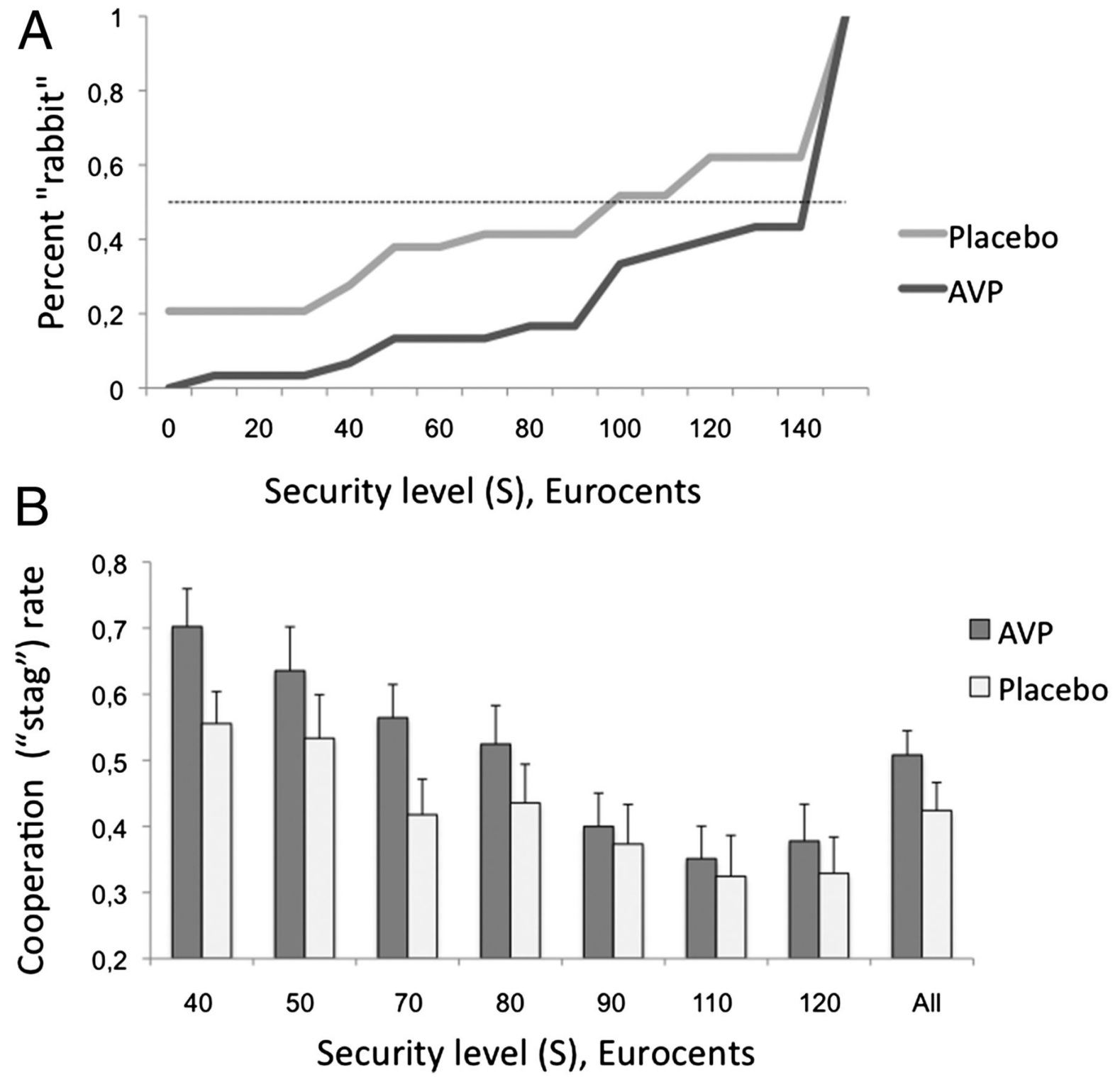
Experiment 1 (Behavior)

Fifty-nine healthy adult males received nasal spray with either 40 International Units (IU) of AVP or placebo, 30 min before the task (19). We presented subjects with a Stag hunt game matrix (Fig. 1*B*) and asked them to choose (22) the minimal security level at which they would switch from the safe Rabbit action to the risky Stag action (we used the abstract labels "A" and "B" rather than Stag and Rabbit to avoid possible framing effects). Then, each subject was randomly matched with a partner, and a value of *S* (between 0 and 150) was realized at random. Based on the drawn *S* value and the subjects' switching points, we determined whether the subjects chose the Stag or Rabbit action and calculated their corresponding payoffs.

The hypothesis that AVP increases cooperative behavior implies that subjects under intranasal AVP would be willing to cooperate even when the security level associated with the Rabbit choice were greater—that is, they would choose a higher switching point than the placebo group. In line with this hypothesis, the AVP group switched from Rabbit to Stag at a higher security level on average [Fig. 2*A*; one-way ANOVA, $F(1,57) = 5.522$, $P < 0.023$].

Fig. 2.

We use cookies on this site to enhance your user experience. By using this website, you are giving your consent for us to set cookies. Find out more
CONTINUE



AVP increased cooperative behavior. (A) AVP subjects switched from Stag to Rabbit at a higher security level (S-value), indicating higher cooperation rates and reduced aversion to social risk under AVP treatment. Note that, as all subjects switched from Stag to Rabbit, the cumulative distribution was equivalent to the rate of Rabbit choice at any given security level. (B) Mean cooperation rates of both groups in experiment 2 (fMRI). AVP subjects were significantly more cooperative; the effect increased when the incentive to cooperate was high.

Several factors might underlie the increased cooperation rates following AVP treatment. First, because Stag payoffs have higher variance (i.e., they are riskier), AVP might simply increase the willingness to accept risky financial payoffs, even absent social context. To test this alternative hypothesis, the Stag hunt game was followed by a task in which the same subjects (still under AVP influence) chose between lotteries with different financial risk levels and no social interdependency (*SI Appendix*, 1.1.3). In this task, where payoffs were kept constant relative to the Stag hunt game but the risk associated with the different options was not socially generated (i.e., was not associated with uncertainty regarding the partner's actions), the AVP group was *not* more likely to take riskier actions than the placebo group [*SI Appendix*, Table S5; one-way ANOVA, $F(1,56) = 0.339$, $P = 0.563$].

Second, AVP-induced cooperation could be mediated by changes in social preferences (i.e., caring about the welfare of one's partner) (23, 24). To measure social preference, the same subjects took part in another task where they chose between five possible monetary

We use cookies on this site to enhance your user experience. By using this website, you are giving your consent for us to set cookies. Find out more
CONTINUE

social preference [*SI Appendix*, Fig. S2; one-way ANOVA, $F(1,57) = 1.242$, $P = 0.270$].

Third, AVP could also influence one's beliefs regarding his partner's actions. We elicited the subjects' beliefs about the likelihood that their partner chose the cooperative strategy by presenting them 12 Stag hunt games in a pseudorandomized order and asking them to indicate the likelihood (between 0 and 100%) that their partner chose "Stag." Again, we found no reliable AVP effect (*SI Appendix*, Fig. S1 and Table S6).

Fourth, we ruled out the possibility that AVP increased cooperation is due to altered self-reported mood [pleasantness: $F(1,57) = 0.48$, $P = 0.490$; wakefulness: $F(1,57) = 1.27$, $P = 0.264$; calmness: $F(1,57) = 0.24$, $P = 0.629$] (*SI Appendix*, 1.2.5).

Our findings demonstrate that AVP's effects on cooperation are specific to the mutually beneficial social context of the Stag hunt game and are consistent with the notion that AVP increases the expected reward of cooperation, rather than general risk attitudes, social preferences, or beliefs about the partner's actions.

Experiment 2 (Behavior and Brain Imaging)

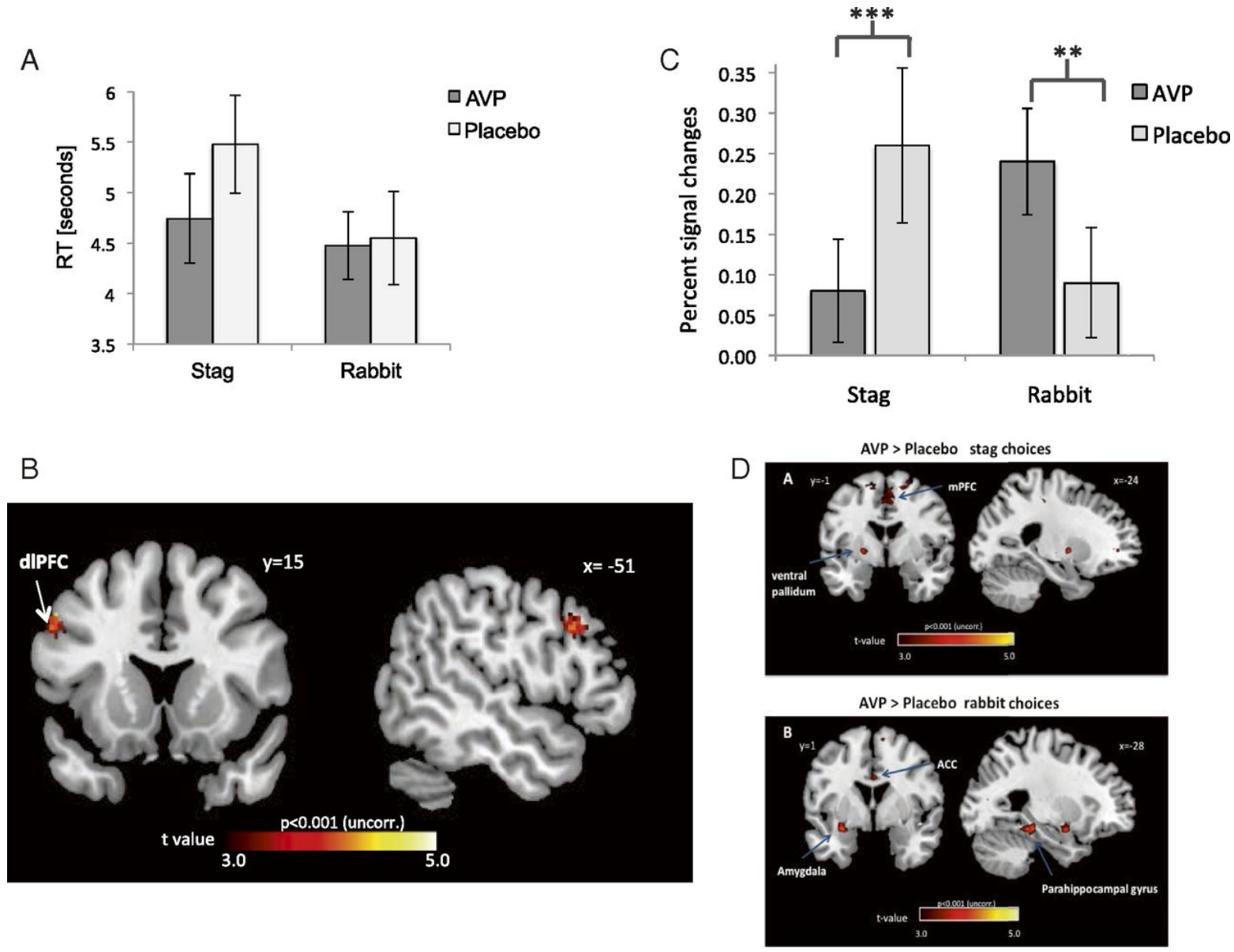
To further investigate the neural mechanisms underlying AVP's effect on risky cooperation, we conducted a second Stag hunt experiment using fMRI. We administered AVP or placebo to 34 healthy adult males in a randomized double-blind regime. Before scanning, each subject was introduced to a fixed partner who, unbeknownst to the subject, was a confederate of the experimenter. Participants played 105 different games derived from seven versions of the Stag hunt game that differed only in their security level (*Materials and Methods* and *SI Appendix*, Tables S8 and S9). Subjects made their choices by pressing a button and received no feedback about their partner's actions and payoffs, to disable the complicated effects of learning and reputation building.

A mixed-effect logistic model with subjects' decisions as the dependent variable (*SI Appendix*, Table S10) revealed significant effects of security level ($z = -6.925$, $P < 0.001$), drug ($z = 2.571$, $P < 0.01$), and a drug × security level interaction ($z = -2.103$, $P < 0.03$). These results replicate the behavioral results of study 1 (Fig. 2*B*). Furthermore, a mixed-effect model with players' response time (RT) as the dependent variable resulted in a significant drug × choice interaction. The AVP group had a significantly faster RT when choosing Stag compared with Rabbit ($t = -2.144$, $P < 0.03$) (Fig. 3*A* and *SI Appendix*, Table S11).

Fig. 3.

We use cookies on this site to enhance your user experience. By using this website, you are giving your consent for us to set cookies. Find out more

CONTINUE



AVP caused choice-dependent changes in brain activity. (A) Mean response times across action choices. The placebo group was significantly slower when choosing Stag compared with Rabbit (paired-sample t = 2.62, P < 0.01); the effect disappeared in the AVP group (t = −2.144, P < 0.03). (B) A cluster in the left dlPFC showed increased BOLD activation for the contrast Stag > Rabbit choices in the placebo group relative to the AVP group. (C) A region-of-interest analysis in the area shown in B revealed a significant drug × choice interaction. AVP decreased the BOLD signal (percentage signal change, y axis) in the left dlPFC during Stag choices (t = −2.34, P < 0.05) and increased activity in the latter brain region during Rabbit choices (t = 1.87, P < 0.05), relative to the placebo group. This finding indicates that cooperation (choosing Stag) required less mental effort under AVP treatment, in accordance with the behavioral and RT data. (D) Intranasal AVP enhanced the left dlPFC functional connectivity with (i) the left ventral pallidum, the cingulate gyrus, and the medial and superior frontal gyrus during Stag choices and (ii) increased the left dlPFC functional connectivity with the left parahippocampal gyrus, left amygdala, and ACC during Rabbit choices.

We estimated cooperation-related brain activity using a general linear model (GLM) on functional MR images measuring blood oxygen level-dependent activity (BOLD signal). The GLM included dummy regressors for actual Stag and Rabbit choices and nuisance parameters controlling for head motion (*Materials and Methods*). We calculated the contrast between BOLD activity from Stag vs. Rabbit choices at the first (subject) level and entered resulting *t*-contrasts of all subjects into two-sample *t* tests by comparing AVP group vs. placebo group and vice versa.

This between-group comparison revealed increased activation of the left dorsolateral prefrontal cortex (dlPFC) (Fig. 3*B*) in the placebo group relative to the AVP group when making Stag vs. Rabbit choices. An additional region-of-interest analysis of the cluster showed that the AVP group had left dlPFC activity that was reduced during Stag choices ($t = -2.34$, $P < 0.05$) and heightened during Rabbit choices ($t = 1.87$, $P < 0.05$) (Fig. 3*C*), relative to the placebo group. Using β-series correlation based functional connectivity analyses (*Materials and Methods*), we

We use cookies on this site to enhance your user experience. By using this website, you are giving your consent for us to set cookies. Find out more
CONTINUE

Table S15 and Figs. S4 and S5). We found that under AVP influence the left dlPFC showed

increased functional connectivity with the left VP, the cingulate gyrus, and the medial and superior frontal gyrus during Stag choices and with the left parahippocampal gyrus, left amygdala, and anterior cingulate cortex (ACC) during Rabbit choices (Fig. 3*D* and *SI Appendix*, Tables S13 and S14).

Discussion

The current study investigated the effect of intranasally administered AVP on mutually beneficial cooperative behavior in humans. We provide, to our knowledge, the first evidence that AVP increases risky cooperative behavior and that this effect is specific to the mutually beneficial context of the Stag hunt game and not mediated by general risk attitudes, a change in beliefs, social preferences, or mood.

Although we acknowledge that inferring cognitive processes from fMRI activation patterns is not straightforward (26), our fMRI investigation suggests a possible neural mechanism underlying AVP-induced cooperation. Given the established links between increased left dlPFC activation and cognitive control (27–29) and the known role of the left dlPFC in risk integration (meta analysis) (30), high-level planning, and behavioral inhibition (31), we speculate that AVP administration has reduced the need to recruit this cortical region to inhibit the choice of the noncooperative Rabbit action during cooperation.

Indeed, Rabbit seems to be the default appetitive choice in the placebo group (as evident by the choice frequencies), but under AVP treatment cooperation (Stag) becomes the default strategy. As a consequence, choosing it requires less processing time (as indicated by the reaction times) and less cognitive control (as indicated by reduced left dlPFC activity) following AVP treatment. Moreover, because under AVP treatment Rabbit is no longer the default

(31), conflict (ACC) (32), and emotional vigilance (amygdala and parahippocampus gyrus) (33).

The enhanced functional coupling of the left dlPFC and VP during cooperative choices in the AVP group provides further support for the hypothesis that AVP increases the intrinsic expected reward associated with cooperation. The VP is part of the basal ganglia reward circuitry and has a high density of both AVP V1a and dopamine receptors (14). Recently, scholars have suggested that the interaction of the dopaminergic and vasopressinergic systems in the VP encodes the rewarding properties of social interactions, facilitating social recognition and pair bonding in several mammalian species (15, 34). Our data suggest that the vasopressinergic/dopaminergic interactions in the reward system (14) are evolutionarily stable across species and might be linked to a broader range of social behaviors as established by animal research so far.

Furthermore, it has been argued that cooperation is intrinsically rewarding for humans (11–13)

We use cookies on this site to enhance your user experience. By using this website, you are giving your consent for us to set cookies. Find out more
CONTINUE

accordance with our finding that AVP facilitates mutually beneficial risky cooperation by

reducing aversion to social risk, without changing general aversion to risky financial outcomes or social preference.

Our experiments delineated AVP's role in mutually beneficial cooperation in an environment where altruistic concerns, reputation building, and strategically learning the partner's intentions and beliefs did not play a role (35, 36). Most importantly, AVP did not change behavior in the social preferences control task, which rules out AVP working purely through increased altruistic one-sided prosociality. However, previous work has shown that AVP administration increased cooperation following a cooperative gesture by one's partner in a repeated prisoner's dilemma (37), where cooperation might be mutually beneficial (38)—in line with the interdependent hypothesis (5). We believe that further investigation of AVP's role in social behavior would benefit from using computational theories that distinguish the different mechanism through which AVP may foster cooperative behavior in repeated interactions (35), particularly belief inference (33, 36, 39) and inferential limits in disorder (40).

In summary, we combined insights and methodologies from biology, anthropology, and economics and documented a biological cause increasing cooperation—a behavioral feature that largely distinguishes humans from other species and is ubiquitous across human societies. Our fMRI results hint for an AVP interaction with cortical control and dopaminergic reward pathways in the VP. Further investigation of this interaction may have clinical implications for treatment of neuropsychiatric disorders associated with social malfunction. Social anxiety and attention deficit disorders, for example, are characterized by low dopamine levels (41–44), and animal research has reported that AVP stimulates dopamine release in the reward system (15, 34). This suggests that intranasally administered AVP may also have the potential to promote dopamine release in human patients suffering from social disorders and thus may have substantial public health significance.

Finally, we highlight that human AVP research is in its early days. We hope that future investigations will further examine whether the effects are generalized to females (45) and other subject populations and allow translating decades of fruitful animal research into biologically informed theories of social behavior in humans.

Materials and Methods

Experiment 1 (Behavior).

Subjects.

Fifty-nine healthy adult males [age = 19–32, mean = 24.3, SD = 2.9; AVP: mean = 24.4, SD = 2.9; placebo: mean = 24.1, SD = 2.9/$t(57) = -0.501$, $P = 0.618$] participated in this study. Exclusion criteria were psychiatric or neurological disorders, kidney disease, cardiovascular problems, asthma, or migraine. All subjects were right-handed. All sessions took place between 8:00 AM and 6:00 PM. Experiments were approved by the ethical committee of the University of

We use cookies on this site to enhance your user experience. By using this website, you are giving your consent for us to set cookies. Find out more
CONTINUE

Experimental procedures and materials.